

Predictive N_T Research Diagnostic

Project 111 · Six-Component COSMO-SAC-2010

Frozen report generated from the completed background-job snapshot. This document is not a pilot-validated process design, does not establish theoretical stages, and must not be used as release evidence.

Run identity

Job ID	1acd1a68-51a0-4877-94fc-0f199eb9f467
Created	31/8/2026, 10:22:31 am
Started	31/8/2026, 10:22:42 am
Completed	31/8/2026, 10:37:05 am
Trials	10 / 10
Component order	SAT / MONO / DI / POLY / PA / NMP
Classification	SIX_COMPONENT_COSMO_SAC_2010_RESEARCH_DIAGNOSTIC

Final verdict

Diagnostic Predictive N_T	—
Established theoretical stages	—
Job status	completed
Pilot validated	false
Release eligible	false
Calibration required	true
Sulfur prediction	NOT_CALCULABLE
Sulfur calibration	—
Overall product acceptance	false
Acceptance status	RESEARCH_DIAGNOSTIC_NOT_RELEASE_ELIGIBLE

Interpretation: all ten requested stage counts were evaluated, but no trial simultaneously passed the numerical, thermodynamic, balance, and saved product-target gates. Therefore Predictive N_T and established theoretical stages remain null.

Frozen Stage 1 design basis

Project reference	111
RRBO grade	SN300
Feed rate	4000 L/h
Operating temperature	30 °C
Operating pressure	2.0
Solvent/oil mass ratio	0.75
Maximum stages	10
Phase configuration	nmp-continuous-rrbo-dispersed
Feed sulfur	3500 ppm
Target raffinate sulfur	1000 ppm
Minimum RRBO recovery	95%
Maximum raffinate NMP	0.5 wt%
Minimum raffinate saturates	90 wt%
Target total aromatics	5 wt%
Target polar aromatics	0.1 wt%

Feed composition (mass basis)

SAT	85 wt%
MONO	7 wt%
DI	4 wt%
POLY	2 wt%
PA	2 wt%
NMP	0 wt%

Fresh modeled solvent is pure NMP at the saved solvent/oil ratio.
Commercial NMP purity and water specifications are audit-only and do not create additional modeled components.

Trial detail — 1 stage

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	2.6534e-14
Overall mole balance residual	2.0122792321330962e-16,1.0408340855860843e-17,3.469446951953614e-18,0,8.673617379884035e-19,-1.1102230246251565e-16
Overall mass balance residual	3.4276118054954454e-14,1.251020120829338e-15,4.933452951716433e-16,0,3.517845736933367e-16,-1.1005762967641886e-14
Multistart product relative difference	4.3943e-11
Acceptance blockers	[object Object]

Product metrics

Sat Loss Pct	5.42394
Nmp In Total Raffinate Wt	7.57282
Component Extraction Pct / DI	49.11813
Component Extraction Pct / PA	66.73647
Component Extraction Pct / MONO	47.99824
Component Extraction Pct / POLY	52.02643
Raffinate Component Mass / DI	2.03527
Raffinate Component Mass / PA	0.66527
Raffinate Component Mass / NMP	7.18467
Raffinate Component Mass / SAT	80.38965
Raffinate Component Mass / MONO	3.64012
Raffinate Component Mass / POLY	0.95947
Raffinate Saturates Wt Nmp Free	91.67504
Nmp Free Hydrocarbon Recovery Pct	87.68979
Raffinate Polar Aromatics Wt Nmp Free	0.75866
Raffinate Total Aromatics Wt Nmp Free	7.5663

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	7.57282
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	87.68979
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	91.67504
Maximum Raffinate Polar Aromatics Wt / Status	FAIL
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.75866
Maximum Raffinate Total Aromatics Wt / Status	FAIL
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	7.5663

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 5.754% MONO 4.193% DI 2.452% POLY 1.299% PA 1.666% NMP 84.636%

Final Raffinate

SAT 84.733% MONO 3.837% DI 2.145% POLY 1.011% PA 0.701% NMP 7.573%

Trial detail — 2 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	6.3838e-14
Overall mole balance residual	7.632783294297951e-17,6.938893903907228e-18,3.469446951953614e-18,4.336808689942018e-18,1.734723475976807e-18,0
Overall mass balance residual	1.3001286158775827e-14,8.340134138862254e-16,4.933452951716433e-16,8.77122159625987e-16,7.035691473866734e-16,0
Multistart product relative difference	0.00055
Acceptance blockers	[object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	6.13864
Nmp In Total Raffinate Wt	7.15911
Component Extraction Pct / DI	66.28829
Component Extraction Pct / PA	86.42899
Component Extraction Pct / MONO	65.14316
Component Extraction Pct / POLY	70.14165
Raffinate Component Mass / DI	1.34847
Raffinate Component Mass / PA	0.27142
Raffinate Component Mass / NMP	6.51124
Raffinate Component Mass / SAT	79.78216
Raffinate Component Mass / MONO	2.43998
Raffinate Component Mass / POLY	0.59717
Raffinate Saturates Wt Nmp Free	94.48475
Nmp Free Hydrocarbon Recovery Pct	84.43919
Raffinate Polar Aromatics Wt Nmp Free	0.32144
Raffinate Total Aromatics Wt Nmp Free	5.19381

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	7.15911
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	84.43919
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	94.48475
Maximum Raffinate Polar Aromatics Wt / Status	FAIL
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.32144
Maximum Raffinate Total Aromatics Wt / Status	FAIL
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	5.19381

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.208% MONO 5.425% DI 3.155% POLY 1.669% PA 2.057% NMP 81.486%

Final Raffinate

SAT 87.72% MONO 2.683% DI 1.483% POLY 0.657% PA 0.298% NMP 7.159%

Trial detail — 3 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	2.7756e-14
Overall mole balance residual	5.551115123125783e-17,6.938893903907228e-18,-6.938893903907228e-18,-1.734723475976807e-18,-4.215378046623641e-16,0
Overall mass balance residual	9.455480842746056e-15,8.340134138862254e-16,-9.866905903432866e-16,-3.508488638503948e-16,-1.7096730281496165e-13,0
Multistart product relative difference	0.00068
Acceptance blockers	[object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	6.52504
Nmp In Total Raffinate Wt	6.94714
Component Extraction Pct / DI	75.24136
Component Extraction Pct / PA	94.1343
Component Extraction Pct / MONO	74.08004
Component Extraction Pct / POLY	79.45622
Raffinate Component Mass / DI	0.99035
Raffinate Component Mass / PA	0.11731
Raffinate Component Mass / NMP	6.18069
Raffinate Component Mass / SAT	79.45371
Raffinate Component Mass / MONO	1.8144
Raffinate Component Mass / POLY	0.41088
Raffinate Saturates Wt Nmp Free	95.97407
Nmp Free Hydrocarbon Recovery Pct	82.78665
Raffinate Polar Aromatics Wt Nmp Free	0.14171
Raffinate Total Aromatics Wt Nmp Free	3.88422

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.94714
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	82.78665
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	95.97407
Maximum Raffinate Polar Aromatics Wt / Status	FAIL
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.14171
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	3.88422

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.447% MONO 6.027% DI 3.498% POLY 1.847% PA 2.188% NMP 79.992%

Final Raffinate

SAT 89.307% MONO 2.039% DI 1.113% POLY 0.462% PA 0.132% NMP 6.947%

Trial detail — 4 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	4.4020e-14
Overall mole balance residual	1.3877787807814457e-17,-1.3877787807814457e-17,0,-3.469446951953614e-18,1.734723475976807e-18,0
Overall mass balance residual	2.363870210686514e-15,-1.6680268277724509e-15,0,-7.016977277007896e-16,7.035691473866734e-16,0
Multistart product relative difference	0.0005
Acceptance blockers	[object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	6.76188
Nmp In Total Raffinate Wt	6.82041
Component Extraction Pct / DI	80.69055
Component Extraction Pct / PA	97.40417
Component Extraction Pct / MONO	79.46832
Component Extraction Pct / POLY	85.02488
Raffinate Component Mass / DI	0.77238
Raffinate Component Mass / PA	0.05192
Raffinate Component Mass / NMP	5.98845
Raffinate Component Mass / SAT	79.25241
Raffinate Component Mass / MONO	1.43722
Raffinate Component Mass / POLY	0.2995
Raffinate Saturates Wt Nmp Free	96.86969
Nmp Free Hydrocarbon Recovery Pct	81.81342
Raffinate Polar Aromatics Wt Nmp Free	0.06346
Raffinate Total Aromatics Wt Nmp Free	3.06685

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.82041
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	81.81342
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	96.86969
Maximum Raffinate Polar Aromatics Wt / Status	PASS
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.06346
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	3.06685

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.591% MONO 6.379% DI 3.701% POLY 1.95% PA 2.234% NMP 79.143%

Final Raffinate

SAT 90.263% MONO 1.637% DI 0.88% POLY 0.341% PA 0.059% NMP 6.82%

Trial detail — 5 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	1.0577e-13
Overall mole balance residual	-3.885780586188048e-16,-6.938893903907228e-18,-3.469446951953614e-16,-1.6132928326584306e-16,-2.612667027168669e-13,5.551115123125783e-16
Overall mass balance residual	-6.618836589922239e-14,-8.340134138862254e-16,-4.933452951716433e-14,-3.2628944338086715e-14,-1.0596454928790688e-10,5.502881483820943e-14
Multistart product relative difference	0.00053
Acceptance blockers	[object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	6.91972
Nmp In Total Raffinate Wt	6.73706
Component Extraction Pct / DI	84.33361
Component Extraction Pct / PA	98.83968
Component Extraction Pct / MONO	83.02726
Component Extraction Pct / POLY	88.66886
Raffinate Component Mass / DI	0.62666
Raffinate Component Mass / PA	0.02321
Raffinate Component Mass / NMP	5.86442
Raffinate Component Mass / SAT	79.11824
Raffinate Component Mass / MONO	1.18809
Raffinate Component Mass / POLY	0.22662
Raffinate Saturates Wt Nmp Free	97.45688
Nmp Free Hydrocarbon Recovery Pct	81.18282
Raffinate Polar Aromatics Wt Nmp Free	0.02859
Raffinate Total Aromatics Wt Nmp Free	2.51454

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.73706
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	81.18282
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	97.45688
Maximum Raffinate Polar Aromatics Wt / Status	PASS
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.02859
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	2.51454

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.687% MONO 6.608% DI 3.835% POLY 2.016% PA 2.248% NMP 78.605%

Final Raffinate

SAT 90.891% MONO 1.365% DI 0.72% POLY 0.26% PA 0.027% NMP 6.737%

Trial detail — 6 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	8.6153e-14
Overall mole balance residual	0,2.0816681711721685e-17,0,0,-1.4849232954361469e-15,0
Overall mass balance residual	0,2.502040241658676e-15,0,0,-6.022551901629924e-13,0
Multistart product relative difference	0.0004
Acceptance blockers	[object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	7.0315
Nmp In Total Raffinate Wt	6.67852
Component Extraction Pct / DI	86.92965
Component Extraction Pct / PA	99.47922
Component Extraction Pct / MONO	85.5326
Component Extraction Pct / POLY	91.20169
Raffinate Component Mass / DI	0.52281
Raffinate Component Mass / PA	0.01042
Raffinate Component Mass / NMP	5.7785
Raffinate Component Mass / SAT	79.02323
Raffinate Component Mass / MONO	1.01272
Raffinate Component Mass / POLY	0.17597
Raffinate Saturates Wt Nmp Free	97.86747
Nmp Free Hydrocarbon Recovery Pct	80.74514
Raffinate Polar Aromatics Wt Nmp Free	0.0129
Raffinate Total Aromatics Wt Nmp Free	2.11963

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.67852
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	80.74514
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	97.86747
Maximum Raffinate Polar Aromatics Wt / Status	PASS
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.0129
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	2.11963

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.755% MONO 6.767% DI 3.93% POLY 2.062% PA 2.249% NMP 78.237%

Final Raffinate

SAT 91.331% MONO 1.17% DI 0.604% POLY 0.203% PA 0.012% NMP 6.679%

Trial detail — 7 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	true
Solver message	`gtol` termination condition is satisfied.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	2.8556e-13
Overall mole balance residual	-4.579669976578771e-16,-9.71445146547012e-17,-4.891920202254596e-16,-2.42861286636753e-16,-7.576881830284599e-13,5.551115123125783e-16
Overall mass balance residual	-7.800771695265496e-14,-1.1676187794407157e-14,-6.956168661920171e-14,-4.911884093905527e-14,-3.0730317327268273e-10,5.502881483820943e-14
Multistart product relative difference	0.00041
Acceptance blockers	[object Object]; [object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	7.11435
Nmp In Total Raffinate Wt	6.63541
Component Extraction Pct / DI	88.86727
Component Extraction Pct / PA	99.76591
Component Extraction Pct / MONO	87.38144
Component Extraction Pct / POLY	93.03927
Raffinate Component Mass / DI	0.44531
Raffinate Component Mass / PA	0.00468
Raffinate Component Mass / NMP	5.71581
Raffinate Component Mass / SAT	78.9528
Raffinate Component Mass / MONO	0.8833
Raffinate Component Mass / POLY	0.13921
Raffinate Saturates Wt Nmp Free	98.1691
Nmp Free Hydrocarbon Recovery Pct	80.4253
Raffinate Polar Aromatics Wt Nmp Free	0.00582
Raffinate Total Aromatics Wt Nmp Free	1.82508

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.63541
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	80.4253
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	98.1691
Maximum Raffinate Polar Aromatics Wt / Status	PASS
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.00582
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	1.82508

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.805% MONO 6.884% DI 4% POLY 2.094% PA 2.245% NMP 77.971%

Final Raffinate

SAT 91.655% MONO 1.025% DI 0.517% POLY 0.162% PA 0.005% NMP 6.635%

Trial detail — 8 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	false
Solver message	The maximum number of function evaluations is exceeded.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	8.9986e-5
Overall mole balance residual	-0.000003665604923,-0.0004686992249768457,-0.00028916101467326913,-0.00018200536202238327,-0.00020329116303118965,0.000019785816224
Overall mass balance residual	-0.0006243800815080395,-0.05633483464686699,-0.04111785771959632,-0.036810693672244224,-0.0824508299021899,0.0019613897266801596
Multistart product relative difference	6.5715e-5
Acceptance blockers	[object Object]; [object Object]; [object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	7.20153
Nmp In Total Raffinate Wt	6.61576
Component Extraction Pct / DI	89.58663
Component Extraction Pct / PA	99.76615
Component Extraction Pct / MONO	88.43031
Component Extraction Pct / POLY	93.1109
Raffinate Component Mass / DI	0.41653
Raffinate Component Mass / PA	0.00468
Raffinate Component Mass / NMP	5.6851
Raffinate Component Mass / SAT	78.8787
Raffinate Component Mass / MONO	0.80988
Raffinate Component Mass / POLY	0.13778
Raffinate Saturates Wt Nmp Free	98.29419
Nmp Free Hydrocarbon Recovery Pct	80.24757
Raffinate Polar Aromatics Wt Nmp Free	0.00583
Raffinate Total Aromatics Wt Nmp Free	1.69998

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.61576
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	80.24757
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	98.29419
Maximum Raffinate Polar Aromatics Wt / Status	PASS
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.00583
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	1.69998

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.857% MONO 6.996% DI 4.06% POLY 2.127% PA 2.327% NMP 77.633%

Final Raffinate

SAT 91.791% MONO 0.942% DI 0.485% POLY 0.16% PA 0.005% NMP 6.616%

Trial detail — 9 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	false
Solver message	The maximum number of function evaluations is exceeded.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	0.00131
Overall mole balance residual	-0.0010506630052797422,-0.003908675836292723,-0.0011209325436500536,-0.0005993511721271456,-0.0009527763854784033,0.00031799435212043203
Overall mass balance residual	-0.17896447287172385,-0.46979938346736755,-0.15939335700266105,-0.12121913417341847,-0.38642704642233083,0.03152312991948576
Multistart product relative difference	0.00106
Acceptance blockers	[object Object]; [object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	6.98022
Nmp In Total Raffinate Wt	6.71937
Component Extraction Pct / DI	83.70918
Component Extraction Pct / PA	80.21003
Component Extraction Pct / MONO	86.56709
Component Extraction Pct / POLY	75.27861
Raffinate Component Mass / DI	0.65163
Raffinate Component Mass / PA	0.3958
Raffinate Component Mass / NMP	5.87429
Raffinate Component Mass / SAT	79.06681
Raffinate Component Mass / MONO	0.9403
Raffinate Component Mass / POLY	0.49443
Raffinate Saturates Wt Nmp Free	96.95623
Nmp Free Hydrocarbon Recovery Pct	81.54897
Raffinate Polar Aromatics Wt Nmp Free	0.48535
Raffinate Total Aromatics Wt Nmp Free	2.55842

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.71937
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	81.54897
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	96.95623
Maximum Raffinate Polar Aromatics Wt / Status	FAIL
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.48535
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	2.55842

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.878% MONO 7.348% DI 3.947% POLY 1.831% PA 2.24% NMP 77.755%

Final Raffinate

SAT 90.441% MONO 1.076% DI 0.745% POLY 0.566% PA 0.453% NMP 6.719%

Trial detail — 10 stages

Research status	RESEARCH_DIAGNOSTIC_NOT_ACCEPTED
Solver success	false
Solver message	The maximum number of function evaluations is exceeded.
Numerical acceptance	false
All calculable targets pass	false
Maximum scaled equation residual	0.00126
Overall mole balance residual	-0.00016986531606451305,-0.004135861722012843,-0.0006390958755246763,-0.0006188071170463456,-0.0010268589348612451,0.000095867624329
Overall mass balance residual	-0.02893397463878562,-0.49710576381561167,-0.09087758012156995,-0.1251541107068936,-0.41647344680102377,0.00950346305415401
Multistart product relative difference	0.00184
Acceptance blockers	[object Object]; [object Object]; [object Object]; [object Object]

Product metrics

Sat Loss Pct	7.17635
Nmp In Total Raffinate Wt	6.70733
Component Extraction Pct / DI	83.98517
Component Extraction Pct / PA	80.70036
Component Extraction Pct / MONO	87.25708
Component Extraction Pct / POLY	75.80337
Raffinate Component Mass / DI	0.64059
Raffinate Component Mass / PA	0.38599
Raffinate Component Mass / NMP	5.8453
Raffinate Component Mass / SAT	78.90011
Raffinate Component Mass / MONO	0.892
Raffinate Component Mass / POLY	0.48393
Raffinate Saturates Wt Nmp Free	97.04496
Nmp Free Hydrocarbon Recovery Pct	81.30263
Raffinate Polar Aromatics Wt Nmp Free	0.47476
Raffinate Total Aromatics Wt Nmp Free	2.48028

Target compliance

Maximum Nmp Raffinate Wt / Status	FAIL
Maximum Nmp Raffinate Wt / Target	0.5
Maximum Nmp Raffinate Wt / Calculated	6.70733
Target Raffinate Sulfur Ppm / Status	NOT_CALCULABLE
Target Raffinate Sulfur Ppm / Target	1000
Target Raffinate Sulfur Ppm / Calculated	—
Minimum Nmp Free Recovery Pct / Status	FAIL
Minimum Nmp Free Recovery Pct / Target	95
Minimum Nmp Free Recovery Pct / Calculated	81.30263
Minimum Raffinate Saturates Wt / Status	PASS
Minimum Raffinate Saturates Wt / Target	90
Minimum Raffinate Saturates Wt / Calculated	97.04496
Maximum Raffinate Polar Aromatics Wt / Status	FAIL
Maximum Raffinate Polar Aromatics Wt / Target	0.1
Maximum Raffinate Polar Aromatics Wt / Calculated	0.47476
Maximum Raffinate Total Aromatics Wt / Status	PASS
Maximum Raffinate Total Aromatics Wt / Target	5
Maximum Raffinate Total Aromatics Wt / Calculated	2.48028

Boundary streams — mass fractions

Oil Feed

SAT 85% MONO 7% DI 4% POLY 2% PA 2% NMP 0%

Fresh Nmp

SAT 0% MONO 0% DI 0% POLY 0% PA 0% NMP 100%

Final Extract

SAT 6.886% MONO 7.421% DI 3.877% POLY 1.844% PA 2.281% NMP 77.69%

Final Raffinate

SAT 90.536% MONO 1.024% DI 0.735% POLY 0.555% PA 0.443% NMP 6.707%

Governance and provenance

Model SHA-256	9666ff67194102ca9c3dc139695bead8ff133bfe24c2d786ce1deca49456196
Engine SHA-256	5d9ab380d8dcadaaed3a8ada30dec216700f8215f8649549491c456a9672057d
Input SHA-256	93486a655df80b5d6c61d2e3400fbe8b9610697ab50edab0bf893424199197df
Stage 1 immutable SHA-256	—
Six-component basis SHA-256	7683442c0419659d83320cfa4433fe8cf298f285f596c4676cf39bb5042838fc
Six-component binding SHA-256	d71161802f9294719416b209739f7918c27cda87a716162fe9308d948271049a
Execution mode	USER_SAVED_STAGE1_RESEARCH_DIAGNOSTIC
Model identity	COSMO-SAC-2010 + PROJECT_NMP_LLE_RESIDUAL
Diagnostic selection basis	FIRST_STAGE_WITH_NUMERICAL_ACCEPTANCE_AND_ALL_CALCULABLE_PRODUCT_TARGETS_PASS; RESEARCH_DIAGNOSTIC_ONLY_NOT_GOVERNED_OR_ACCEPTED
Monotonic sequence	false

Controlled limitations

- Sulfur removal is NOT CALCULABLE from this six-component aromatic-transfer diagnostic.
- The PA representative is non-sulfur-bearing and is not a sulfur surrogate.
- No trial is an accepted or release-eligible ECR design result.
- Predictive N_T is null because no trial satisfied all qualification gates.
- Calibration and direct matching NMP/RRBO evidence remain required.

